

Dhaniel Bolosan

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EDUCATION

University of Hawai'i at Mānoa

Bachelor of Science in Computer Science

Honolulu, HI

Aug. 2024 - May 2026

University of Hawai'i Maui College

Associate of Science in Natural Science — Information and Computer Sciences

Kahului, HI

Aug. 2022 - May 2024

EXPERIENCE

Software Engineer Intern

June 2026 – July 2026

Black Sand Solutions

Kihei, HI

- Engineered a geospatial intelligence dashboard and backend data pipeline to query and render live data streams across desktop and mobile platforms.
- Built interactive Space Domain Awareness simulations, including ML-based satellite thermal anomaly detection for missile launches, 5G mesh network visualizations, and a game-theoretic drone swarm engine.

Undergraduate AI Security Researcher

Jan. 2026 – May 2026

University of Hawai'i at Mānoa

Honolulu, HI

- Investigated RAG model vulnerabilities in partnership with NIWC by designing methods to evaluate data retrieval integrity.
- Developed exploits using targeted vector manipulation and invisible text injection in PDF documents to manipulate LLM data retrieval, demonstrating critical vulnerabilities in RAG pipelines.

Full-Stack Software Engineer Intern

Jan. 2026 – May 2026

Blockchain in Paradise

Honolulu, HI

- Developed a decentralized supply chain tracker on Base Sepolia Testnet using Solidity smart contracts to verify the origin of local Hawaiian coffee products.
- Built a full-stack Web3 application using Next.js and Vercel, integrating QR code generation, 3D mapping, NFT creation, and IPFS-based asset storage.

Backend Cloud Engineer Intern

Aug. 2025 – Dec. 2025

University of Hawai'i Office of Government Relations

Honolulu, HI

- Designed and deployed a cost-efficient FastAPI backend on Google Cloud, powering an AI bill summarization tool and automated daily notification system serving UH staff and officials.
- Shipped production-ready tools integrated into the office's internal website, replacing manual legislative analysis and notification workflows.

PROJECTS

AI Academic Assistant | REST APIs, Next.js, Neon Postgres, Tailwind CSS

Oct. 2025 – Nov. 2025

- Developed a full-stack AI chat interface using Next.js and Tailwind CSS, enabling students to interactively query and explore university course data through natural language.
- Engineered a RAG pipeline using OpenAI APIs and Neon Postgres to enable semantic search across structured academic data, reducing time spent manually navigating course catalogs.
- Enhanced usability by integrating a custom text-to-speech tool and graphical course diagrams.

Inventory Management App | React Bootstrap, Typescript, Prisma, Playwright

Aug. 2025 – Dec. 2025

- Built a full-stack pantry and recipe management application with a 9-person team, delivering real-time stock tracking, automated shopping lists, and recipe generation.
- Maintained codebase stability by architecting CI workflows and Playwright end-to-end tests, ensuring reliable deployments across a distributed team.
- Collaborated across the entire codebase using TypeScript, Prisma ORM, and React Bootstrap, contributing to both frontend components and backend data modeling.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL (PostgreSQL), HTML/CSS, Solidity

Frameworks: React, Svelte, Next.js, Node.js, Tailwind CSS, FastAPI, REST APIs

Developer Tools: Git, GitHub Actions, CI/CD, Docker, Linux/Unix, Google Cloud Platform, Playwright

Libraries: LangChain, OpenAI SDK, Anthropic SDK, Ollama, Redis, Drizzle ORM, Prisma ORM